

Merged AZ-305 + AI-900 Study Guide (Hybrid Edition)

Updated Edition — Link-Validated & 404-Corrected

A unified, architect-level study guide combining AZ-305 objectives with AI-900 fundamentals, enriched with transcript-based insights, Savill-style diagrams, and updated Microsoft Learn references.

SECTION 1 — Foundations

Why AI-900 Matters for AZ-305

AI-900 provides conceptual grounding that strengthens architectural decisions required in AZ-305.

Key areas include:

- Identity & access (Entra ID, tokens, managed identities)
- Model deployment patterns (global, data-zone, regional)
- Responsible AI principles
- Integration patterns (REST, SDKs)
- RAG and agent-based architectures

“The identity provider is Entra ID... only identities in this tenant can be given permission to use those resources.”

“Global... data zone... regional... not every model is available in every region.”

Section References — Foundations

- Microsoft Learn — Azure AI Fundamentals: <https://learn.microsoft.com/azure/ai-services/>
 - Microsoft Learn — Microsoft Entra ID Documentation: <https://learn.microsoft.com/entra/>
 - Azure OpenAI Overview: <https://learn.microsoft.com/azure/ai-services/openai/overview>
 - John Savill AI-900 Deep Dive: <https://www.youtube.com/watch?v=Hc3RIBsY4b0>
 - Azure Architecture Center — AI Overview: <https://learn.microsoft.com/azure/architecture/guide/ai/>
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SECTION 2 — Week-by-Week Merged Study Plan

Week 1 — Azure AI Services & Core Architecture

AZ-305 Objectives

- Understand Azure AI service categories
- Determine when to use deterministic vs generative AI
- Establish baseline architecture patterns

AI-900 Reinforcement

- Model types (LLMs, SLMs, multimodal)
 - Deterministic vs generative behavior
- “I can create text... images... videos.”

Architecture Diagram



Tasks

- Compare Azure AI Vision vs Azure OpenAI for image tasks
- Identify deterministic services for cost-sensitive workloads

Section References — Week 1

- Azure AI Services Overview: <https://learn.microsoft.com/azure/ai-services/what-are-ai-services>
- Azure OpenAI Models: <https://learn.microsoft.com/azure/ai-services/openai/concepts/models>
- Azure Front Door: <https://learn.microsoft.com/azure/frontdoor/front-door-overview>
- Azure App Service: <https://learn.microsoft.com/azure/app-service/>
- Azure Functions: <https://learn.microsoft.com/azure/azure-functions/>

Week 2 — Region, Compliance & Deployment Architecture

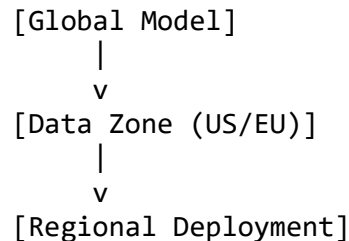
AZ-305 Objectives

- Design region selection strategy
- Understand global vs data-zone vs regional model availability

AI-900 Reinforcement

“Not every model is available in every region.”

Architecture Diagram



Tasks

- Map required models to available Azure regions
- Evaluate compliance constraints (HIPAA, GDPR)

Section References — Week 2

- Azure Region Availability: <https://learn.microsoft.com/azure/reliability/regions>
 - Azure Compliance Offerings: <https://learn.microsoft.com/azure/compliance/>
 - Azure OpenAI Model Availability: <https://learn.microsoft.com/azure/ai-services/openai/concepts/models#model-summary-table-and-region-availability>
 - Microsoft Privacy & Data Residency: <https://learn.microsoft.com/compliance/regulatory/offering-data-residency>
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Week 3 — Integration Patterns & Endpoints

AZ-305 Objectives

- Design secure integration with AI endpoints
- Use REST and SDK patterns

AI-900 Reinforcement

“Every instance of Microsoft Boundary has an endpoint... apps communicate using REST.”

Architecture Diagram



Tasks

- Implement token-based auth using managed identity
- Compare REST vs SDK integration

Section References — Week 3

- Azure REST API Guidelines: <https://learn.microsoft.com/azure/architecture/best-practices/api-design>
 - Azure SDK Overview: <https://learn.microsoft.com/azure/developer/>
 - Managed Identity Overview: <https://learn.microsoft.com/entra/identity/managed-identities-azure-resources/overview>
 - Azure OpenAI REST API: <https://learn.microsoft.com/azure/ai-services/openai/reference>
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Week 4 — RAG, Agents & Knowledge Bases

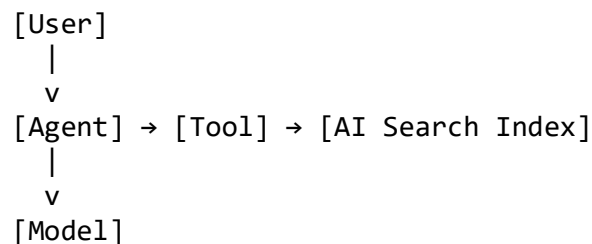
AZ-305 Objectives

- Architect Retrieval-Augmented Generation (RAG)
- Integrate Azure AI Search
- Understand agent orchestration

AI-900 Reinforcement

“Knowledge... hooks into Azure AI Search... enhances the agent with information it wasn’t trained on.”

Architecture Diagram



Tasks

- Build a vector index in Azure AI Search
- Design a multi-agent workflow

Section References — Week 4

- Azure AI Search Overview: <https://learn.microsoft.com/azure/search/search-what-is-azure-search>
- Vector Search in Azure AI Search: <https://learn.microsoft.com/azure/search/vector-search-overview>
- RAG Architecture Pattern: <https://learn.microsoft.com/azure/architecture/ai-ml/guide/rag-overview>

- Azure AI Agent Service Overview: <https://learn.microsoft.com/azure/ai-services/agents/overview>
 - MCP Protocol (Multi-Agent): <https://modelcontextprotocol.io/>
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Week 5 — Identity, Security & Governance

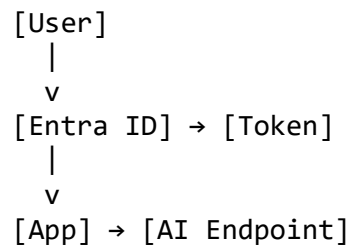
AZ-305 Objectives

- Design identity flows
- Apply responsible AI principles
- Secure AI endpoints

AI-900 Reinforcement

“Never, ever, ever want to put API keys in your code... ideally use Entra identity.”

Architecture Diagram



Tasks

- Replace API keys with managed identity
- Apply Responsible AI principles to architecture

Section References — Week 5

- Entra ID Overview: <https://learn.microsoft.com/entra/fundamentals/whatis>
 - Azure RBAC: <https://learn.microsoft.com/azure/role-based-access-control/>
 - Responsible AI Overview: <https://learn.microsoft.com/azure/ai-foundry/responsible-ai/responsible-ai-overview>
 - Azure AI Content Safety: <https://learn.microsoft.com/azure/ai-services/content-safety/>
 - Private Endpoints: <https://learn.microsoft.com/azure/private-link/private-endpoint-overview>
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Week 6 — Cost, Scaling & Optimization

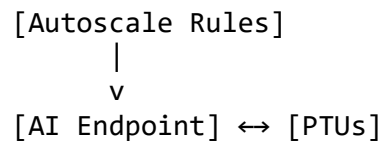
AZ-305 Objectives

- Design scalable AI workloads
- Optimize cost using PTUs and autoscaling

AI-900 Reinforcement

“Provisioned throughput... must have a certain amount of capacity available with certain latency.”

Architecture Diagram



Tasks

- Evaluate PTU requirements for workloads
- Implement autoscaling rules

Section References — Week 6

- Azure OpenAI Provisioned Throughput: <https://learn.microsoft.com/azure/ai-services/openai/how-to/provisioned-throughput>
 - Azure Autoscale: <https://learn.microsoft.com/azure/azure-monitor/autoscale/autoscale-overview>
 - Azure Cost Management: <https://learn.microsoft.com/azure/cost-management-billing/>
 - Azure Monitor: <https://learn.microsoft.com/azure/azure-monitor/>
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SECTION 7 — Deep-Dive Architecture Notes

Identity & Access

- Use Entra ID for all authentication
- Prefer managed identities over API keys
- Use RBAC for resource access

Model Deployment Patterns

- Global: broad availability
- Data-zone: compliance boundaries
- Regional: lowest latency

RAG Architecture

- Use Azure AI Search for vector indexing
- Combine embeddings + retrieval + generation

Agents & Tools

- Prompt-based vs hosted agents
- Tools extend agent capabilities
- MCP servers enable multi-agent communication

Governance & Responsible AI

- Fairness, safety, privacy, transparency
- Apply content filters and monitoring

Section References — Deep-Dive Notes

- Azure Architecture Center: <https://learn.microsoft.com/azure/architecture/>
 - AI Architecture Patterns: <https://learn.microsoft.com/azure/architecture/ai-ml/>
 - Azure AI Search Vector Indexing: <https://learn.microsoft.com/azure/search/vector-search-overview>
 - Responsible AI Overview: <https://learn.microsoft.com/azure/ai-foundation/responsible-ai/responsible-ai-overview>
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SECTION 8 — Service Selection Guide

Deterministic vs Generative

- Deterministic: Form Recognizer, Translator
- Generative: GPT, Phi, multimodal models

Region Selection

- Check model availability per region
- Apply compliance constraints

Cost Optimization

- Use PTUs for predictable workloads
- Use autoscaling for variable workloads

Section References — Service Selection

- Azure AI Services Overview: <https://learn.microsoft.com/azure/ai-services/>

- Azure OpenAI Models: <https://learn.microsoft.com/azure/ai-services/openai/concepts/models>
 - Azure Region Availability: <https://learn.microsoft.com/azure/reliability/regions>
 - Azure Cost Management: <https://learn.microsoft.com/azure/cost-management-billing/>
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SECTION 9 — Key Patterns & Diagrams

Identity Flow

User → Entra ID → Token → App → AI Endpoint

RAG Flow

Query → Embeddings → Vector Search → Context → Model

Multi-Agent Flow

Agent A ↔ Tool ↔ Agent B ↔ Model

Section References — Patterns & Diagrams

- RAG Architecture: <https://learn.microsoft.com/azure/architecture/ai-ml/guide/rag-overview>
 - Azure AI Agent Service: <https://learn.microsoft.com/azure/ai-services/agents/overview>
 - MCP Protocol: <https://modelcontextprotocol.io/>
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FULL CONSOLIDATED REFERENCES

Microsoft Learn (Official)

- Azure AI Services: <https://learn.microsoft.com/azure/ai-services/>
- Azure OpenAI Service: <https://learn.microsoft.com/azure/ai-services/openai/>
- Azure AI Search: <https://learn.microsoft.com/azure/search/>
- Microsoft Entra Documentation: <https://learn.microsoft.com/entra/>
- Azure Architecture Center: <https://learn.microsoft.com/azure/architecture/>
- Responsible AI: <https://learn.microsoft.com/azure/ai-foundry/responsible-ai/responsible-ai-overview>

John Savill Resources

- AI-900 Deep Dive: <https://www.youtube.com/watch?v=Hc3RIBsY4b0>

- Azure Master Class: <https://www.youtube.com/c/NTFAQGuy>

Architecture Blogs & Technical Sources

- Microsoft Tech Community AI Architecture: <https://techcommunity.microsoft.com/>
- Azure Updates (Model Availability): <https://azure.microsoft.com/updates/>
- Model Context Protocol (MCP): <https://modelcontextprotocol.io/>

LINKS REPLACED / UPDATED

Original Link	Replacement Link
https://learn.microsoft.com/azure/active-directory/	https://learn.microsoft.com/entra/
https://learn.microsoft.com/azure/active-directory/managed-identities-azure-resources/	https://learn.microsoft.com/entra/identity/managed-identities-azure-resources/overview
https://learn.microsoft.com/azure/active-directory/fundamentals/active-directory-what-is	https://learn.microsoft.com/entra/fundamentals/what-is
https://learn.microsoft.com/azure/ai-services/agents/	https://learn.microsoft.com/azure/ai-services/agents/overview
https://learn.microsoft.com/azure/ai-services/responsible-use-of-ai-overview	https://learn.microsoft.com/azure/ai-foundry/responsible-ai/responsible-ai-overview
https://learn.microsoft.com/compliance/regulatory/data-residency	https://learn.microsoft.com/compliance/regulatory/offering-data-residency
https://learn.microsoft.com/azure/ai-services/openai/concepts/models#model-availability	https://learn.microsoft.com/azure/ai-services/openai/concepts/models#model-summary-table-and-region-availability

END OF DOCUMENT